Benewake SR18

Datasheet



SR18 is a new-generation LiDAR developed by Benewake for intelligent robotics and industrial automation. Its compact form factor and powerful performance deliver high-resolution sensing across the entire field of view for accurate detection of a wide range of targets.

SR18 supports flexible embedded integration to accommodate a variety of robot designs. It is suitable for mobile robot navigation and obstacle avoidance, UAV obstacle avoidance, perimeter protection, and other applications.

Product Features

Long-range Detection

Ultra-high Angular Resolution

Compact Size

Low Power Consumption

Applications

Robots

UAVs

Intelligent Control

Last-mile Delivery

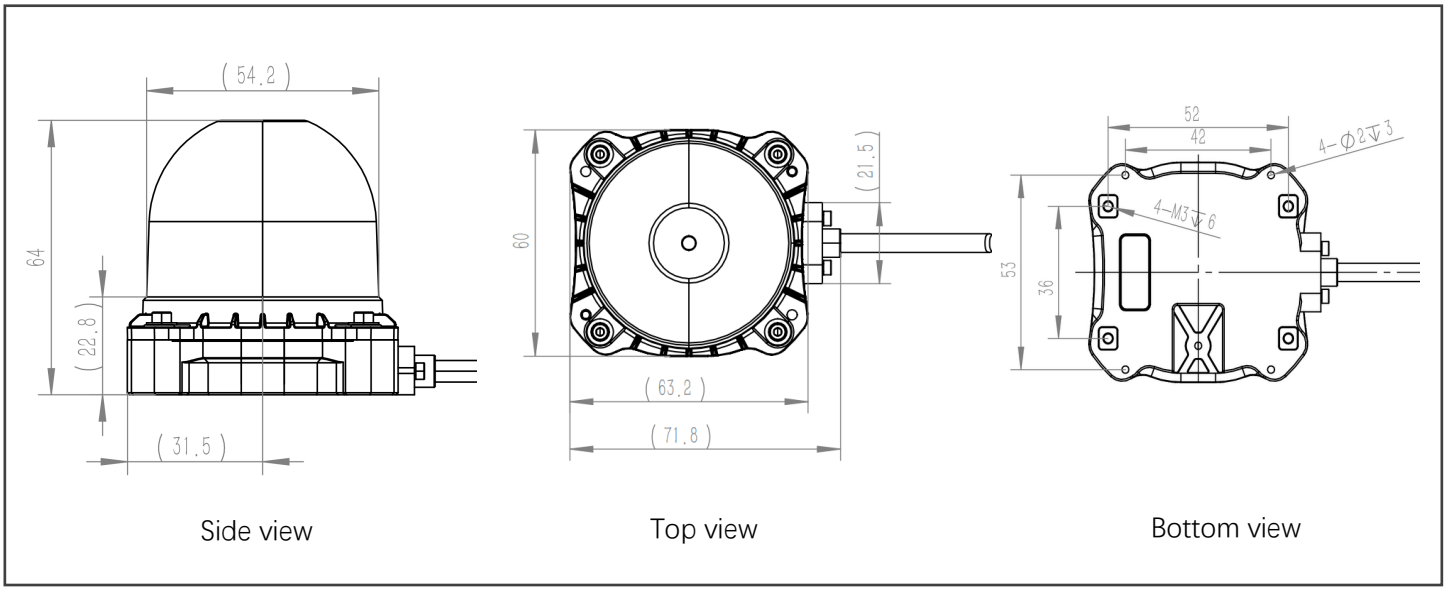
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Technical Specifications

Measurement Performance	
Detection Range	30 m @10% reflectivity
Minimum Detection Range	≤0.1 m
Field of View	360° × 48° (-8° to 40°)
Horizontal Angular Resolution @ 5 Hz	0.6°
Vertical Angular Resolution @ 5 Hz	2.67° (0° to 40°)
	4° (-8° to 0°)
Ranging Accuracy	3 cm @1σ
Point Rate	54 kHz (single return)
Frame Rate	5 Hz (default); 10 Hz customizable
Return Mode	Single return (strongest, farthest, or nearest selectable)
Laser Parameters	
Laser Wavelength	940 nm
Laser Safety Class	Class 1 Eye-safe [IEC 60825-1:2014]
Data Output	
Data Communication	RS485 (baud rate: 312500)
Supported Platforms	Windows, Linux (Ubuntu), and ROS driver
Built-in IMU	200 Hz output (default)
Time Synchronization	PTP, PPS + GPS
Mechanical/Electrical	
Average Power Consumption	≤4 W
Operating Voltage	9–16 V DC
Operating Temperature	-20°C to +60°C
Storage Temperature	-40°C to +85°C
Dimensions (Diameter × Height)	Ø71.8mm × 64 mm (H)
Weight	190g
Ingress Protection	IP66

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2D drawings



Side view

Top view

Bottom view

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